

● HARD

● ALGEBRA

Linear functions

01

10 questions · ~15 min

Q1

ALGEBRA

For the function f , if $f(3x) = x - 6$ for all values of x , what is the value of $f(6)$?

A. -6 B. -4 C. 0 D. 2

For the function f , $fcx = x - 8$ for all values of x , where c is a positive constant. If $f2 = 35$, what is the value of c ?

STUDENT-PRODUCED RESPONSE

.	/	.	.
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.

Q3

GRID-IN ALGEBRA

The equation $h = \frac{9v - 273.15}{5} + 32$ gives the corresponding temperature h , in degrees Fahrenheit, of any substance that has a temperature of v kelvins, where $v > 0$. If a substance has a temperature of 467.33 degrees Fahrenheit, what is the corresponding temperature, in kelvins, of this substance?

STUDENT-PRODUCED RESPONSE

.	/	.	.
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.

Oil and gas production in a certain area dropped from 4 million barrels in 2000 to 1.9 million barrels in 2013. Assuming that the oil and gas production decreased at a constant rate, which of the following linear functions f best models the production, in millions of barrels, t years after the year 2000?

A. $f(t) = \frac{21}{130}t + 4$

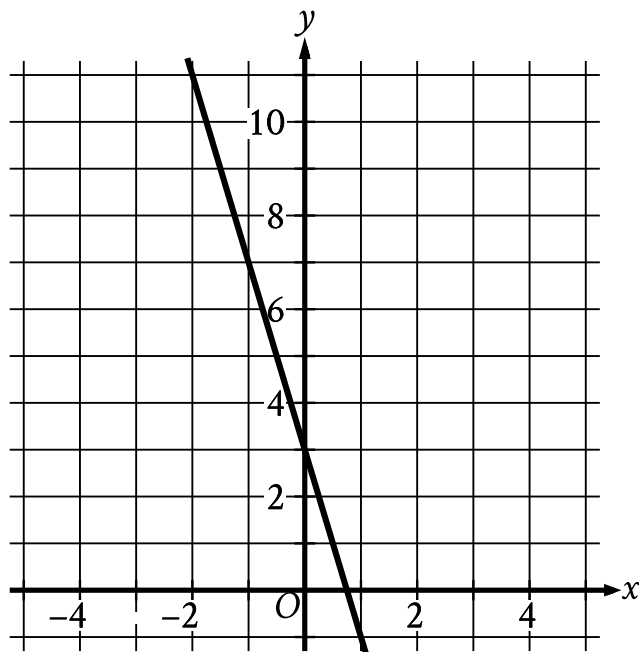
B. $f(t) = \frac{19}{130}t + 4$

C. $f(t) = -\frac{21}{130}t + 4$

D. $f(t) = -\frac{19}{130}t + 4$

An economist modeled the demand Q for a certain product as a linear function of the selling price P . The demand was 20,000 units when the selling price was \$40 per unit, and the demand was 15,000 units when the selling price was \$60 per unit. Based on the model, what is the demand, in units, when the selling price is \$55 per unit?

- A. 16,250
- B. 16,500
- C. 16,750
- D. 17,500



- The line slants sharply down from left to right.
- The line passes through the following points:
 - (negative 1 comma 7)
 - (0 comma 3)

The graph of the linear function $y = fx + 19$ is shown. If c and d are positive constants, which equation could define f ?

- A. $fx = -d - cx$
- B. $fx = d - cx$
- C. $fx = -d + cx$
- D. $fx = d + cx$

x	fx
1	-64
2	0
3	64

For the linear function f , the table shows three values of x and their corresponding values of fx . Function f is defined by $fx = ax + b$, where a and b are constants. What is the value of $a - b$?

- A. -64
- B. 62
- C. 128
- D. 192

For a particular car, the linear function f gives the predicted power, in brake horsepower bhp, for engine speeds between 1,000 revolutions per minute rpm and 6,000 rpm. According to this function, the car's predicted power is 433 bhp at an engine speed of 3,331 rpm and 600 bhp at an engine speed of 4,500 rpm. The equation $f(x) = \frac{1}{7}x - a + 433$ defines f , where x is the engine speed, in rpm, and a is a constant. What is the value of a ?

STUDENT-PRODUCED RESPONSE

.	/.	/.	.
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.

Q9

ALGEBRA

The cost of renting a carpet cleaner is \$ 52 for the first day and \$ 26 for each additional day. Which of the following functions gives the cost Cd , in dollars, of renting the carpet cleaner for d days, where d is a positive integer?

- A. $C(d) = 26d + 26$
- B. $C(d) = 26d + 52$
- C. $C(d) = 52d - 26$
- D. $C(d) = 52d + 78$

Q10

ALGEBRA

$$Fx = \frac{9}{5}x - 273.15 + 32$$

The function F gives the temperature, in degrees Fahrenheit, that corresponds to a temperature of x kelvins. If a temperature increased by 2.10 kelvins, by how much did the temperature increase, in degrees Fahrenheit?

- A. 3.78
- B. 35.78
- C. 487.89
- D. 519.89